



# ELIZA and Pattern Matching

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## 1. What is ELIZA?

- **ELIZA** is one of the earliest natural language processing (NLP) computer programs.
  - Created by **Joseph Weizenbaum** in 1966 at MIT.
  - ELIZA simulates a conversation by using **pattern matching** and **substitution rules**.
  - Famous for mimicking a Rogerian psychotherapist, reflecting user input as questions or statements.
  - ELIZA gave the illusion of understanding, but it had **no real comprehension** or intelligence.
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## 2. Purpose and Significance

- ELIZA demonstrated that simple pattern matching can create convincing dialogue.
  - It raised important questions about the nature of **understanding** in machines.
  - Served as a foundation for later AI chatbots and conversational agents.
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## 3. How ELIZA Works: Pattern Matching

### Core Idea:

- ELIZA processes user input by comparing it to a **set of predefined patterns**.



- Each pattern corresponds to a **response template**.
- When a pattern matches, ELIZA generates a response using that template, sometimes reusing parts of the user's input.

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## Components:

| Component           | Description                                                    |
|---------------------|----------------------------------------------------------------|
| Patterns            | Predefined templates or rules to match user input text.        |
| Keywords            | Important words ELIZA looks for to trigger specific responses. |
| Decomposition Rules | How input sentences are broken down to match patterns.         |
| Reassembly Rules    | How matched fragments are used to form replies.                |
| Substitution        | Replace pronouns or words to make replies more natural.        |

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## Example:

**User Input:** "I am feeling sad today."

- ELIZA might match the pattern: I am \*
- The \* is a wildcard matching the rest: feeling sad today



- Reassembly rule for this pattern might be: "Why do you say you are \*?"
  - ELIZA's response: "Why do you say you are feeling sad today?"
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## 4. Pattern Matching Process in ELIZA

### 1. Keyword Identification:

- ELIZA searches for keywords in the user input in order of priority.

### 2. Decomposition:

- Break the input sentence into parts using the matched pattern.

### 3. Reassembly:

- Insert the extracted parts into a response template.

### 4. Pronoun Substitution:

- Swap first-person and second-person pronouns to maintain conversational flow.

### 5. Response Generation:

- Output the generated response to the user.
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## 5. Example Pattern, Decomposition, and Reassembly Rules

| Keyword | Decomposition<br>Pattern | Reassembly Rule |
|---------|--------------------------|-----------------|
|---------|--------------------------|-----------------|



|         |             |                           |
|---------|-------------|---------------------------|
| I       | I *         | Why do you say you *?     |
| You     | You *       | What makes you think I *? |
| My      | My *        | Tell me more about your * |
| Because | * because * | Is that the real reason?  |

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## 6. Pronoun Substitution Examples

| Original Word | Substituted Word |
|---------------|------------------|
|---------------|------------------|

|     |      |
|-----|------|
| I   | you  |
| me  | you  |
| my  | your |
| am  | are  |
| you | I    |

Example:

User says: "I feel happy."

ELIZA substitutes to respond: "Why do you say you feel happy?"

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## 7. Limitations of ELIZA

- **No understanding:** Purely syntactic pattern matching; no semantic analysis.
- **Rule-based:** Responses limited to predefined patterns; easily predictable.
- **Context Ignorant:** Does not remember past interactions or context.
- **Not intelligent:** Does not truly comprehend user input.

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## 8. Legacy and Influence

- ELIZA inspired future chatbots and dialogue systems.
- Showed how simple rule-based pattern matching can simulate human conversation to some extent.
- Highlighted challenges in natural language understanding.
- Modern AI chatbots use far more complex methods (machine learning, deep learning), but pattern matching remains a core concept in some rule-based systems.

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## 9. Summary Table

### Aspect

### Details

Developed by     Joseph Weizenbaum, 1966

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|                |                                                |
|----------------|------------------------------------------------|
| Purpose        | Simulate conversation via pattern matching     |
| Core technique | Pattern matching with decomposition/reassembly |
| Strengths      | Simple, effective at simulating dialogue       |
| Limitations    | No real understanding, limited flexibility     |
| Legacy         | Foundation for chatbots and conversational AI  |